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DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

DSA06 – DATA HANDLING AND VISUALIZATION

CO3_AT1_Assessment Tool: EXPERIMENT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each experiment, use an appropriate visualization, justify the design, interpret the result, and identify potential visualization bias. R/ggplot2 or an equivalent visualization tool may be used.

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

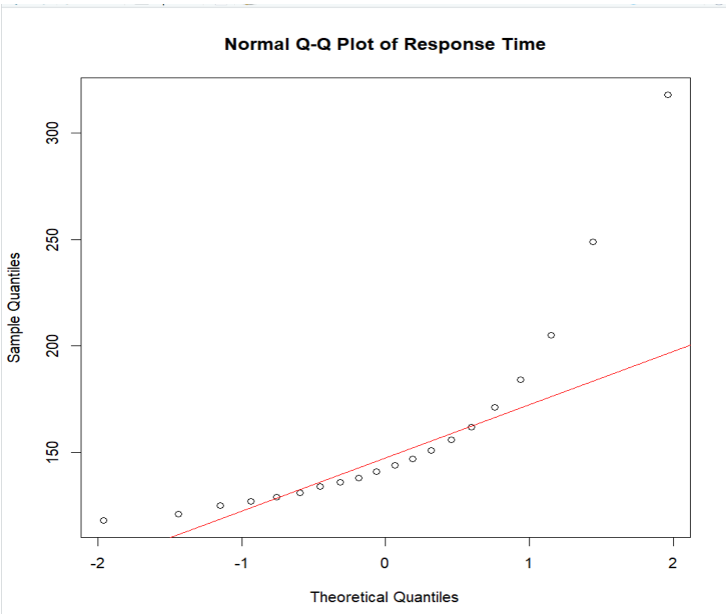
Tasks:

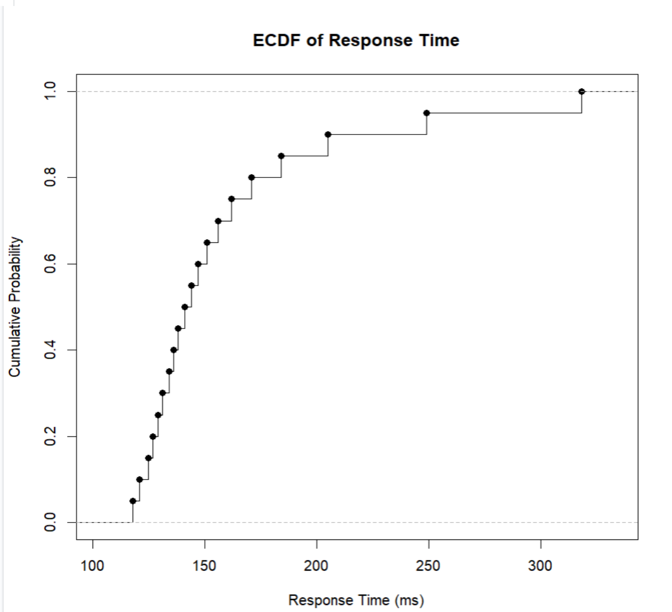
1. **Construct an ECDF and a normal Q-Q plot.**

Code :

```
library(ggplot2)

response<-c(118,121,125,127,129,131,134,136,138,141,144,147,151,156,162,171,184,205,249,318)

data<-data.frame(response)

ggplot(data,aes(x=response))+

stat_ecdf()+

labs(title="ECDF of Response Time",

x="Response Time (ms)",

y="Cumulative Probability")+

theme_minimal()

qqnorm(response,main="Normal Q-Q Plot of Response Time",xlab="Theoretical Quantiles",ylab="Sample Quantiles")

qqline(response,col="red")
```

Output :





2. **Use both plots to evaluate the normality claim.**
   A. The ECDF and normal Q-Q plot indicate that the response-time data is not perfectly normally distributed. The Q-Q plot shows noticeable deviations from the reference straight line, especially at the upper end, while the ECDF shows a longer upper tail. Therefore, the team's claim that the data is approximately normal is not strongly supported.
3. **Identify evidence of skewness and/or extreme observations.**
   A. The data shows positive (right) skewness, because most response times are concentrated at lower values while a few observations extend far into the upper range. The values 205 ms, 249 ms, and 318 ms form the upper tail, with 318 ms being the most extreme observation. The Q-Q plot also shows deviation from the reference line at the upper end.
4. **Recommend percentile-based limits or mean ± standard-deviation limits, with justification.**
   A. Percentile-based limits are recommended for this response-time data. Since the distribution is right-skewed and contains extreme high observations, the mean and standard deviation can be influenced by these values. Percentile-based limits are therefore more suitable for monitoring the response times and identifying unusually high values.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

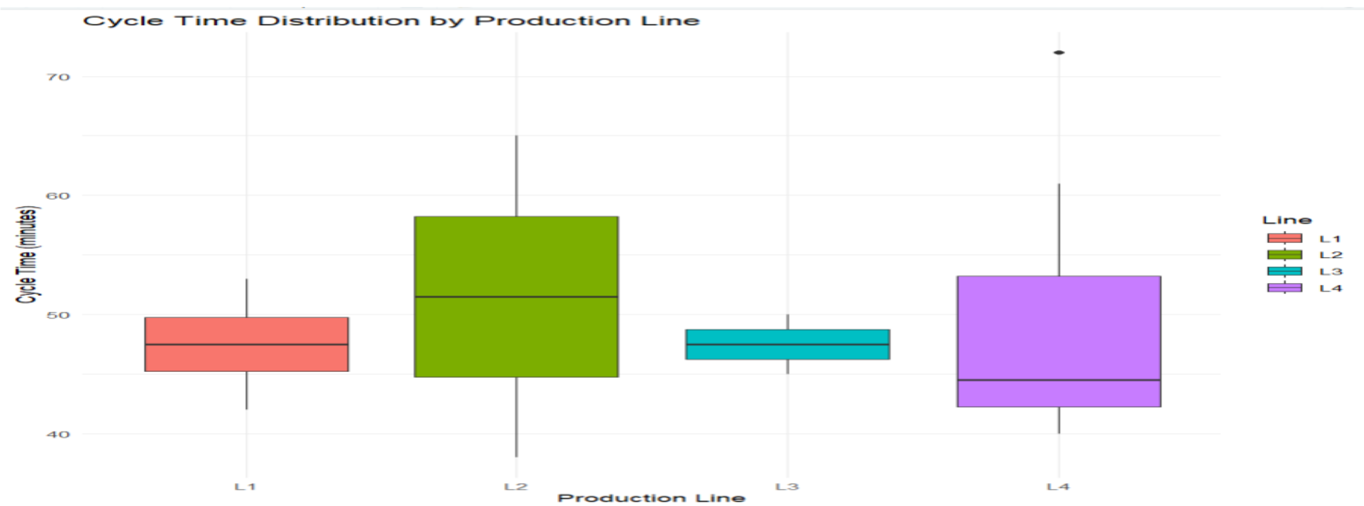
Production Line	Cycle Times (minutes)
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. **Create a visualization comparing all four distributions.**

```
library(ggplot2)

data<-data.frame(

  Line=rep(c("L1","L2","L3","L4"),each=10),

  Cycle_Time=c(

    42,44,45,46,47,48,49,50,51,53,

    38,41,44,47,50,53,56,59,62,65,

    45,46,46,47,47,48,48,49,49,50,

    40,41,42,43,44,45,48,55,61,72

  )

)

ggplot(data,aes(x=Line,y=Cycle_Time,fill=Line))+

  geom_boxplot()+

  labs(title="Cycle Time Distribution by Production Line",x="Production Line",y="Cycle Time (minutes)")+

  theme_minimal()
```

Output :

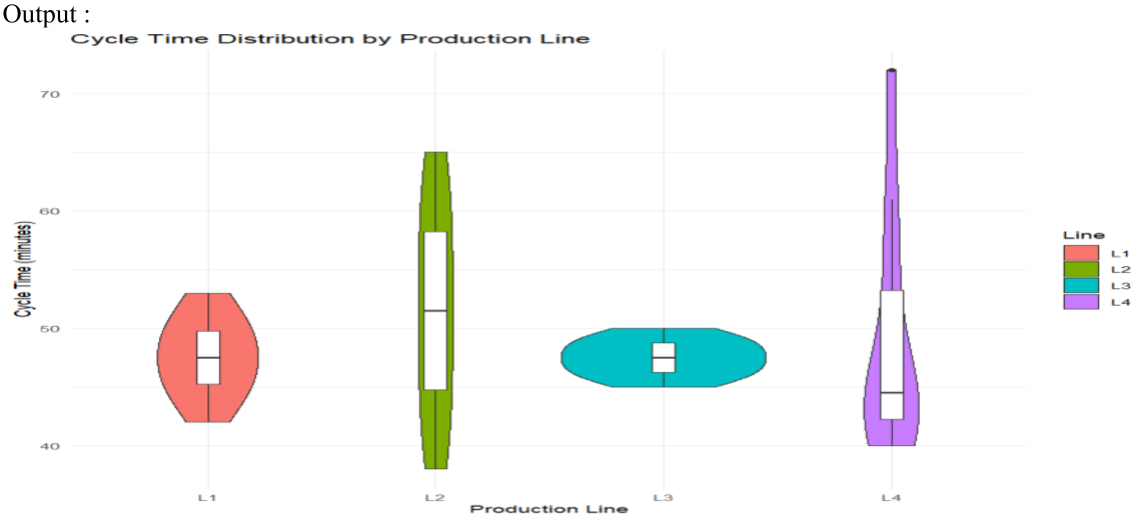


2. **Construct bean plots or an equivalent distribution visualization along the vertical axis.**

Code :

```
library(ggplot2)
data<-data.frame(
  Line=rep(c("L1","L2","L3","L4"),each=10),
Cycle_Time=c(42,44,45,46,47,48,49,50,51,53,38,41,44,47,50,53,56,59,62,65,45,46,46,47,47,48,48,49,49,50,40,41,42,43,44,45,48,55,
61,72)
)
ggplot(data,aes(x=Line,y=Cycle_Time,fill=Line))+
  geom_violin()+

  labs(title="Cycle Time Distribution by Production Line",x="Production Line",y="Cycle Time (minutes)")+
  theme_minimal()
```

Output :



3. **Determine which line has greatest variability and which has the most concentrated distribution.**

- L4 has the greatest variability because its cycle times range widely from 40 to 72 minutes, showing a large spread.
- L3 has the most concentrated distribution because its values are closely grouped between 45 and 50 minutes, showing very little variation.

4. **Explain why comparing only means is insufficient.**
   Comparing only the mean cycle time is insufficient because the mean does not show the variability or distribution shape of the data. Two production lines can have similar means but very different spreads, skewness, or extreme observations.
   Therefore, the distributions should be compared using visualizations such as bean plots/violin plots, which show both the center and variability of the cycle times.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.

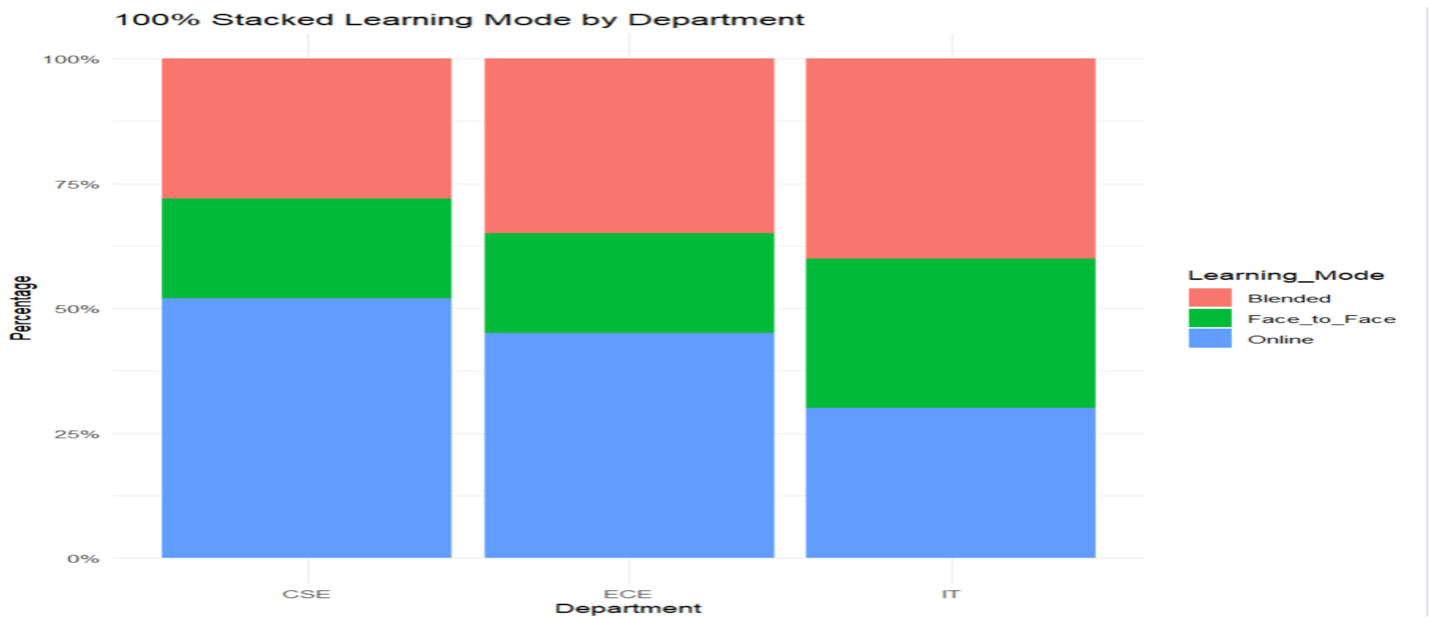
Code :

```
library(ggplot2)

library(tidyr)

data<-data.frame(

Department=c("CSE","ECE","IT"),

Online=c(52,45,30),

Blended=c(28,35,40),

Face_to_Face=c(20,20,30)

)

data_long<-pivot_longer(data,cols=c(Online,Blended,Face_to_Face),names_to="Learning_Mode",values_to="Percentage")

ggplot(data_long,aes(x=Department,y=Percentage,fill=Learning_Mode))+

geom_col(position="dodge")+

labs(title="Learning Mode by Department",x="Department",y="Percentage")+

theme_minimal()


ggplot(data_long,aes(x=Department,y=Percentage,fill=Learning_Mode))+

geom_col(position="fill")+

scale_y_continuous(labels=function(x)paste0(x*100,"%"))+
```

labs(title="100% Stacked Learning Mode by Department",x="Department",y="Percentage")+

theme_minimal()

Output :



2. **Explain which chart better supports comparison of individual proportions and why.**
   The side-by-side bar chart better supports comparison of individual proportions. It is easier to compare the heights of the bars for Online, Blended, and Face-to-Face learning across departments than comparing segments within a stacked chart.
3. **Explain why separate pie charts may be less effective for comparing departments.**
   Separate pie charts are less effective for comparing departments because it is difficult to compare the sizes of individual slices across different pie charts accurately. A side-by-side bar chart or 100% stacked bar chart makes differences in learning-mode proportions easier to compare
4. **Identify one design choice that could make small proportional differences appear larger.**
   A design choice that could make small proportional differences appear larger is using a truncated y-axis in a bar chart instead of starting the axis at zero. This can visually exaggerate small differences between the departments.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	**Subcategory**	**Revenue (₹ lakh)**
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

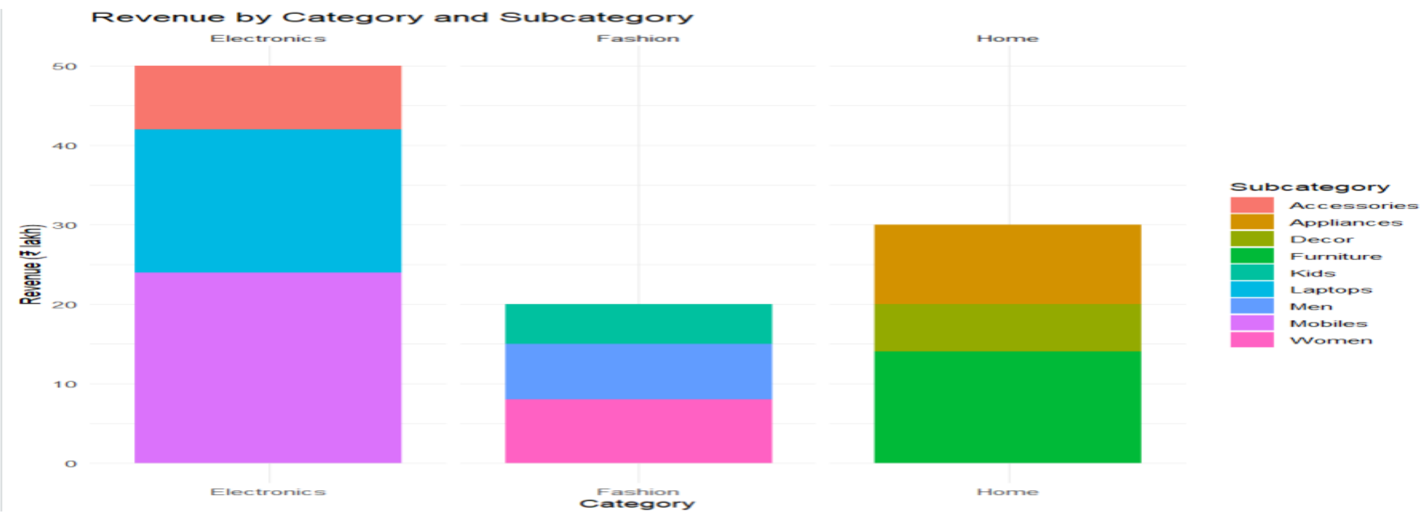
Tasks:

1. **Construct a treemap showing category and subcategory revenue.**

Code :

```
library(ggplot2)

data<-data.frame(

Category=c("Electronics","Electronics","Electronics","Home","Home","Home","Fashion","Fashion","Fashion"),

Subcategory=c("Mobiles","Laptops","Accessories","Furniture","Appliances","Decor","Men","Women","Kids"),

Revenue=c(24,18,8,14,10,6,7,8,5))

ggplot(data,aes(x=Category,y=Revenue,fill=Subcategory))+

geom_col()+

facet_wrap(~Category,scales="free_x")+

labs(title="Revenue by Category and Subcategory",x="Category",y="Revenue (₹ lakh)")+

theme_minimal()
```

Output :

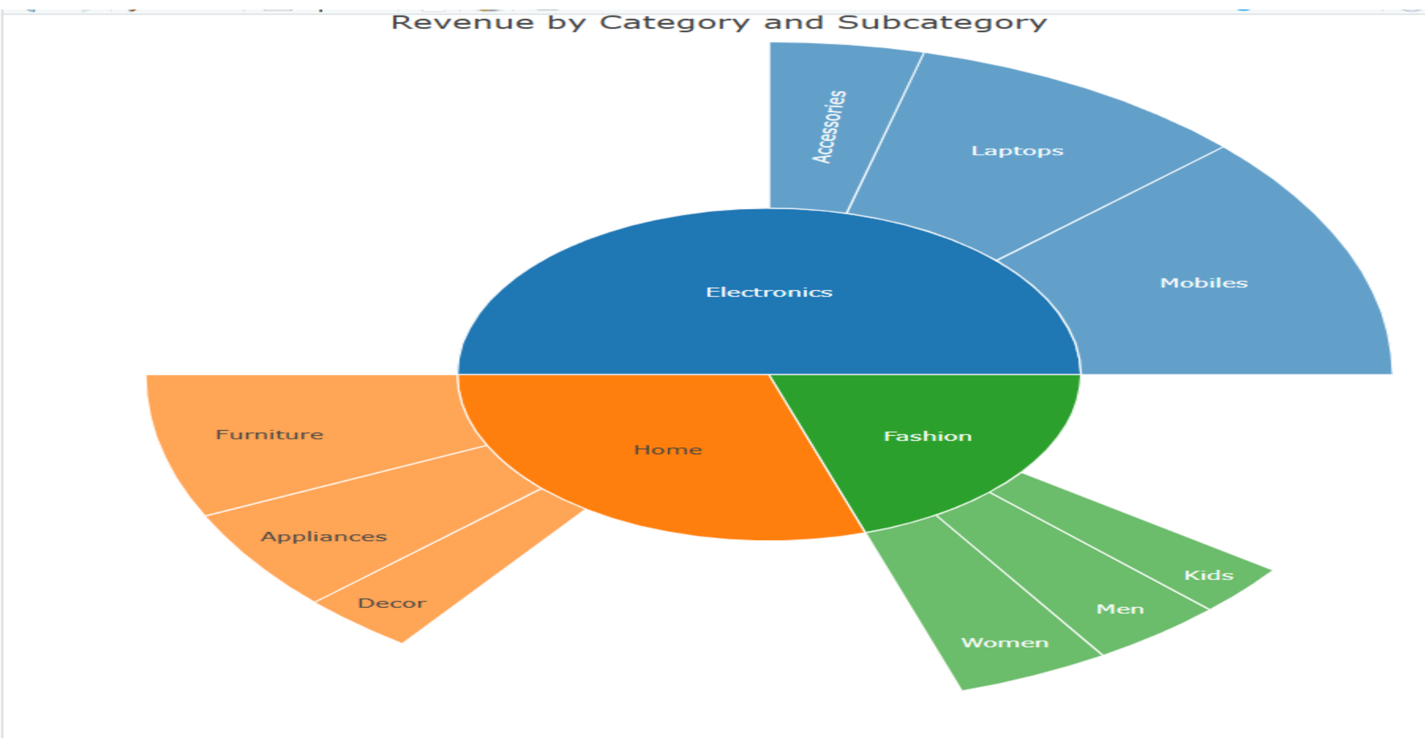


2. **Construct a sunburst chart representing the same hierarchy.**

Code :

```
library(plotly)
data<-data.frame(
Category=c("Electronics","Electronics","Electronics","Home","Home","Home","Fashion","Fashion","Fashion"),
Subcategory=c("Mobiles","Laptops","Accessories","Furniture","Appliances","Decor","Men","Women","Kids"),
Revenue=c(24,18,8,14,10,6,7,8,5)
)
category_total<-aggregate(Revenue~Category,data=data,sum)
labels<-c(category_total$Category,data$Subcategory)
parents<-c("","","",data$Category)
values<-c(category_total$Revenue,data$Revenue)
fig<-plot_ly(
labels=labels,
parents=parents,
values=values,
type="sunburst"
)
fig<-fig%>%layout(title="Revenue by Category and Subcategory")
fig
```

Output :



3. **Determine the largest category and calculate its percentage of total revenue.**

Step 1: Calculate total revenue

- **Electronics:** 24 + 18 + 8 = **50**
- **Home:** 14 + 10 + 6 = **30**
- **Fashion:** 7 + 8 + 5 = **20**

Total revenue = 50 + 30 + 20 = 100

Step 2: Find the largest category

Electronics = 50

Therefore:

Largest category = Electronics

Step 3: Calculate percentage

50/100 * 100 = 50%

4. **Compare treemap and sunburst for identifying nested proportions.**
   A treemap is better for quickly comparing the relative sizes of categories and subcategories because the area of each rectangle represents revenue. A sunburst chart is better for understanding the hierarchical relationship between categories and subcategories. Therefore, the treemap is useful for comparing revenue proportions, while the sunburst is useful for understanding the nested structure.
5. **State one situation in which area encoding could mislead interpretation.**
   Area encoding can mislead when people estimate the size of rectangles based on their width or height rather than their actual area. Small differences in area may also be difficult to judge accurately, especially when rectangles have different shapes. Therefore, a treemap may make some categories appear more or less important than they actually are.

Key competency: Treemaps, sunburst charts, nested proportions, area encoding

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets

Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:

1. **Construct a Sankey diagram showing Origin → Mode → Destination.**

Code :

```
library(networkD3)

links<-data.frame(

source=c(0,0,0,0,1,1,1,1,2,2,2,2,3,3,3,3,4,4),

target=c(3,4,3,4,3,4,3,4,3,4,3,4,5,6,5,6,5,6),

value=c(35,15,25,10,20,10,30,15,12,8,10,10,1,1,1,1,1,1)

)

nodes<-data.frame(

name=c("Residential","Suburban","Rural","Bus","Train","Central","Industrial")

)
```

sankeyNetwork(

Links=links,

Nodes=nodes,

Source="source",

Target="target",

Value="value",

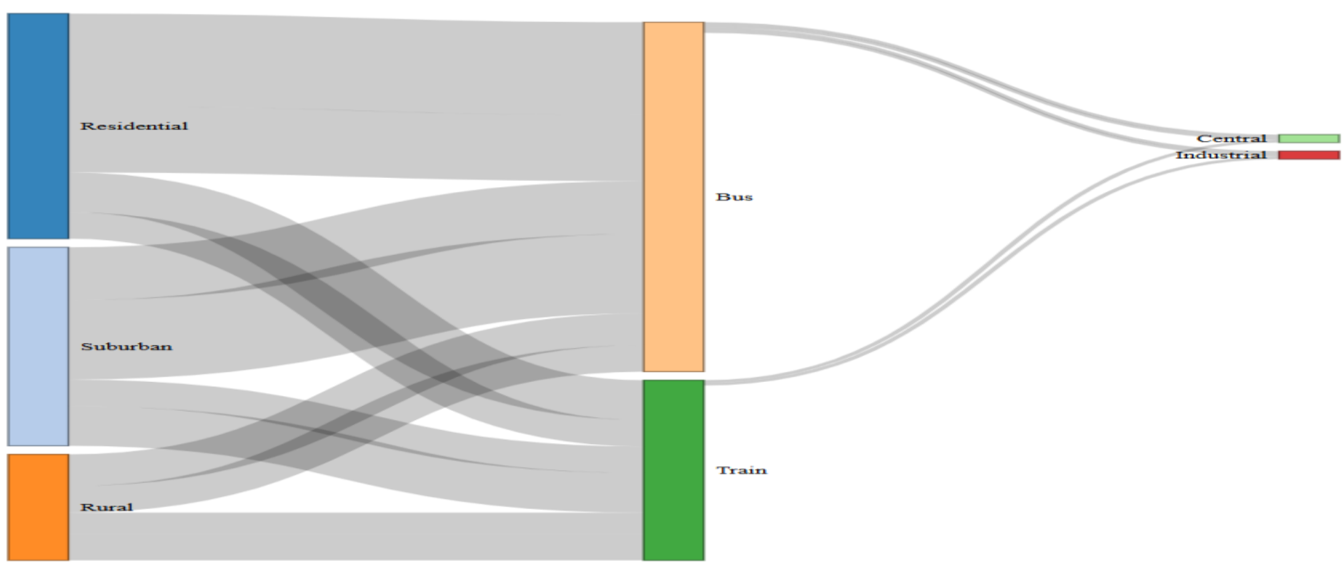
NodeID="name",

fontSize=12,

nodeWidth=30

)

Output :



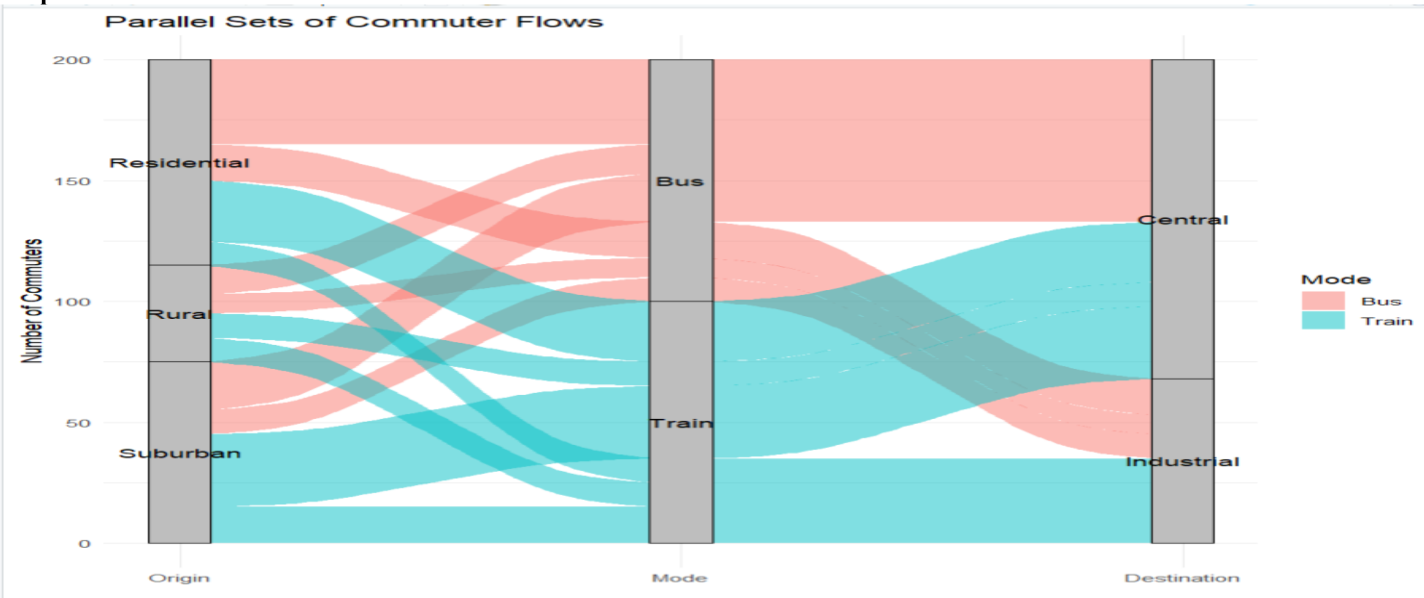
2. **Construct a parallel-sets representation of the same categorical flows.**

Code :

```
library(ggplot2)
library(ggalluvial)
data<-data.frame(
Origin=c("Residential","Residential","Residential","Residential","Suburban","Suburban","Suburban","Suburban","Rural","Rural","Ru
ral","Rural"),
Mode=c("Bus","Bus","Train","Train","Bus","Bus","Train","Train","Bus","Bus","Train","Train"),
Destination=c("Central","Industrial","Central","Industrial","Central","Industrial","Central","Industrial","Central","Industrial","Central
","Industrial"),
Commuters=c(35,15,25,10,20,10,30,15,12,8,10,10)
)
ggplot(data,aes(axis1=Origin,axis2=Mode,axis3=Destination,y=Commuters))+
geom_alluvium(aes(fill=Mode),width=1/12)+
geom_stratum(width=1/8,fill="grey",color="black")+
geom_text(stat="stratum",aes(label=after_stat(stratum)))+
scale_x_discrete(limits=c("Origin","Mode","Destination"),expand=c(.05,.05))+
labs(title="Parallel Sets of Commuter Flows",x="",y="Number of Commuters")+
```

theme_minimal()

Output :



3. **Identify the three largest flows and calculate their percentages of all commuters.**
The three largest commuter flows are Residential → Bus → Central (35 commuters, 17.5%), Suburban → Train → Central (30 commuters, 15%), and Residential → Train → Central (25 commuters, 12.5%). Together, these three flows account for 90 commuters, or 45% of all commuters.
4. **Compare Sankey and parallel sets for flow magnitude and category relationships.**
A Sankey diagram is better for showing the magnitude of flows because the width of each link directly represents the number of commuters. A parallel-sets chart is better for understanding relationships between categorical dimensions such as Origin, Mode, and Destination. Therefore, Sankey is more effective for identifying dominant flows, while parallel sets is useful for comparing category combinations and relationships.
5. **Explain one way poor link widths or category ordering could mislead the viewer.**
If the link widths in a Sankey diagram are not proportional to the actual number of commuters, a smaller flow may appear more important than a larger flow. Similarly, poor ordering of categories can make related flows difficult to follow and may cause the viewer to misunderstand the dominant routes.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

Suggested Evaluation Rubric – Experiment

Criterion	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.